

Lab 9 — Sinusoids & Phasors

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Description

This lab looks at how series RC, RL, and RLC circuits behave when they are driven by a sine wave instead of a DC source. Each circuit is built on the trainer, driven with a 10 V peak-to-peak sine source, and the frequency is swept from 100 Hz to 10 kHz. At each frequency we measure the amplitude of the reactive element voltage and its phase difference relative to the source.

The point is to see impedance behave the way resistance does not. Impedance is a complex number $Z = R + jX$, so the series combination rules from DC still apply, but the reactance X depends on frequency: $Z_C = 1/(j\omega C)$ shrinks as frequency goes up, while $Z_L = j\omega L$ grows. Because the impedances are complex, the element voltages come out shifted in phase from the source, and the voltage divider that sets each element voltage changes as the frequency changes. We express all of the results as phasors and compare the measured magnitude and phase against the calculated values.

Procedure

1. Built a series RC circuit with the 4.7 k Ω resistor and the 0.1 μ F capacitor across the sine wave source on the XK700 trainer.
2. Set the source amplitude to 10 V peak-to-peak and verified it on the oscilloscope. Kept channel 1 on the source and channel 2 on the element being measured.
3. Stepped the source frequency through 100 Hz, 500 Hz, 1 kHz, 5 kHz, and 10 kHz. At each step we rechecked the source amplitude (the generator output sags a little when the load impedance drops) and recorded V_C peak-to-peak and the phase difference ϕ_C between V_C and the source.
4. Measured phase from the time shift between the two zero crossings, $\phi = -360^\circ \Delta t f$, taking the sign from whether the element voltage crossed before or after the source.
5. Rebuilt the circuit as a series RL using the 100 Ω resistor and the 33 mH inductor, and repeated steps 2–4 recording V_L and ϕ_L .
6. Rebuilt the circuit as a series RLC using the 1 k Ω resistor, the 0.33 μ F capacitor, and the 10 mH inductor. Repeated steps 2–4 at each frequency, this time recording both V_C , ϕ_C and V_L , ϕ_L . Moving the scope probe between the capacitor and the inductor required re-referencing ground, so we reordered the series elements as needed to keep the element under test next to the grounded side.

Calculations

All three circuits are a single series loop, so the total impedance is just the sum of the element impedances and each element voltage follows from the impedance divider. With the source taken

as the phase reference, $\mathbf{V}_s = 10\angle 0^\circ$ V (peak-to-peak),

$$\mathbf{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}, \quad \mathbf{Z}_L = j\omega L, \quad \omega = 2\pi f.$$

RC circuit ($R = 4.7\text{ k}\Omega$, $C = 0.1\text{ }\mu\text{F}$):

$$\mathbf{Z}_T = R + \frac{1}{j\omega C}, \quad \mathbf{V}_C = \mathbf{V}_s \frac{\mathbf{Z}_C}{\mathbf{Z}_T}, \quad \mathbf{V}_R = \mathbf{V}_s \frac{R}{\mathbf{Z}_T}.$$

RL circuit ($R = 100\text{ }\Omega$, $L = 33\text{ mH}$):

$$\mathbf{Z}_T = R + j\omega L, \quad \mathbf{V}_L = \mathbf{V}_s \frac{\mathbf{Z}_L}{\mathbf{Z}_T}.$$

RLC circuit ($R = 1\text{ k}\Omega$, $C = 0.33\text{ }\mu\text{F}$, $L = 10\text{ mH}$):

$$\mathbf{Z}_T = R + j\left(\omega L - \frac{1}{\omega C}\right), \quad \mathbf{V}_C = \mathbf{V}_s \frac{\mathbf{Z}_C}{\mathbf{Z}_T}, \quad \mathbf{V}_L = \mathbf{V}_s \frac{\mathbf{Z}_L}{\mathbf{Z}_T}.$$

The series RLC has a resonant frequency where the two reactances cancel:

$$f_0 = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{(10\text{ mH})(0.33\text{ }\mu\text{F})}} = 2.77\text{ kHz},$$

which falls between the 1 kHz and 5 kHz test points.

```
// Lab 9 - Sinusoids & Phasors
// Series RC, RL, RLC: total impedance and element voltage phasors

f = [100 500 1000 5000 10000]; // Hz
Vs = 10; // Vpp, reference phase 0
w = 2*pi*f;

function polarprint(name, z)
    mprintf(" %-4s = %9.3f angle %8.2f deg\n", name, abs(z), atan(imag(z)
    ),real(z))*180/pi);
endfunction

// ----- RC: R = 4.7k, C = 0.1u -----
R = 4700; C = 0.1e-6;
mprintf("\n=== RC circuit: R = 4.7k, C = 0.1u ===\n");
for k = 1:length(f)
    Zc = 1/(i*w(k)*C);
    Zt = R + Zc;
    Vc = Vs*Zc/Zt;
    Vr = Vs*R/Zt;
    mprintf("f = %6.0f Hz\n", f(k));
    polarprint("Zc", Zc); polarprint("Zt", Zt);
    polarprint("Vc", Vc); polarprint("Vr", Vr);
end

// ----- RL: R = 100, L = 33m -----
R = 100; L = 33e-3;
mprintf("\n=== RL circuit: R = 100, L = 33m ===\n");
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for k = 1:length(f)
    Zl = %i*w(k)*L;
    Zt = R + Zl;
    Vl = Vs*Zl/Zt;
    Vr = Vs*R/Zt;
    mprintf("f = %6.0f Hz\n", f(k));
    polarprint("Zl", Zl); polarprint("Zt", Zt);
    polarprint("Vl", Vl); polarprint("Vr", Vr);
end

// ----- RLC: R = 1k, C = 0.33u, L = 10m -----
R = 1000; C = 0.33e-6; L = 10e-3;
f0 = 1/(2*%pi*sqrt(L*C));
mprintf("\n=== RLC circuit: R = 1k, C = 0.33u, L = 10m ===\n");
mprintf("resonant frequency f0 = %.1f Hz\n", f0);
for k = 1:length(f)
    Zc = 1/(%i*w(k)*C);
    Zl = %i*w(k)*L;
    Zt = R + Zc + Zl;
    Vc = Vs*Zc/Zt;
    Vl = Vs*Zl/Zt;
    mprintf("f = %6.0f Hz\n", f(k));
    polarprint("Zl", Zl); polarprint("Zc", Zc); polarprint("Zt", Zt);
    polarprint("Vc", Vc); polarprint("Vl", Vl);
end

```

Listing 1: SciLab program: theoretical impedances and phasor voltages

```

=== RC circuit: R = 4.7k, C = 0.1u ===
f =      100 Hz
  Zc = 15915.494 angle -90.00 deg
  Zt = 16594.968 angle -73.55 deg
  Vc =      9.591 angle -16.45 deg
  Vr =      2.832 angle  73.55 deg
f =      500 Hz
  Zc =  3183.099 angle -90.00 deg
  Zt =  5676.453 angle -34.11 deg
  Vc =      5.608 angle -55.89 deg
  Vr =      8.280 angle  34.11 deg
f =     1000 Hz
  Zc =  1591.549 angle -90.00 deg
  Zt =  4962.160 angle -18.71 deg
  Vc =      3.207 angle -71.29 deg
  Vr =      9.472 angle  18.71 deg
f =     5000 Hz
  Zc =   318.310 angle -90.00 deg
  Zt =  4710.767 angle -3.87 deg
  Vc =      0.676 angle -86.13 deg
  Vr =      9.977 angle   3.87 deg
f =    10000 Hz
  Zc =   159.155 angle -90.00 deg
  Zt =  4702.694 angle -1.94 deg
  Vc =      0.338 angle -88.06 deg
  Vr =      9.994 angle   1.94 deg

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```

=== RL circuit: R = 100, L = 33m ===
f = 100 Hz
  Zl = 20.735 angle 90.00 deg
  Zt = 102.127 angle 11.71 deg
  Vl = 2.030 angle 78.29 deg
  Vr = 9.792 angle -11.71 deg
f = 500 Hz
  Zl = 103.673 angle 90.00 deg
  Zt = 144.042 angle 46.03 deg
  Vl = 7.197 angle 43.97 deg
  Vr = 6.942 angle -46.03 deg
f = 1000 Hz
  Zl = 207.345 angle 90.00 deg
  Zt = 230.200 angle 64.25 deg
  Vl = 9.007 angle 25.75 deg
  Vr = 4.344 angle -64.25 deg
f = 5000 Hz
  Zl = 1036.726 angle 90.00 deg
  Zt = 1041.537 angle 84.49 deg
  Vl = 9.954 angle 5.51 deg
  Vr = 0.960 angle -84.49 deg
f = 10000 Hz
  Zl = 2073.451 angle 90.00 deg
  Zt = 2075.861 angle 87.24 deg
  Vl = 9.988 angle 2.76 deg
  Vr = 0.482 angle -87.24 deg

=== RLC circuit: R = 1k, C = 0.33u, L = 10m ===
resonant frequency f0 = 2770.5 Hz
f = 100 Hz
  Zl = 6.283 angle 90.00 deg
  Zc = 4822.877 angle -90.00 deg
  Zt = 4919.307 angle -78.27 deg
  Vc = 9.804 angle -11.73 deg
  Vl = 0.013 angle 168.27 deg
f = 500 Hz
  Zl = 31.416 angle 90.00 deg
  Zc = 964.575 angle -90.00 deg
  Zt = 1367.767 angle -43.02 deg
  Vc = 7.052 angle -46.98 deg
  Vl = 0.230 angle 133.02 deg
f = 1000 Hz
  Zl = 62.832 angle 90.00 deg
  Zc = 482.288 angle -90.00 deg
  Zt = 1084.409 angle -22.76 deg
  Vc = 4.447 angle -67.24 deg
  Vl = 0.579 angle 112.76 deg
f = 5000 Hz
  Zl = 314.159 angle 90.00 deg
  Zc = 96.458 angle -90.00 deg
  Zt = 1023.423 angle 12.28 deg
  Vc = 0.942 angle -102.28 deg
  Vl = 3.070 angle 77.72 deg

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```

f = 10000 Hz
Zl = 628.319 angle 90.00 deg
Zc = 48.229 angle -90.00 deg
Zt = 1156.073 angle 30.12 deg
Vc = 0.417 angle -120.12 deg
Vl = 5.435 angle 59.88 deg

```

Listing 2: SciLab runtime output

Theoretical Results

Table 1: RC circuit, theoretical ($R = 4.7 \text{ k}\Omega$, $C = 0.1 \text{ }\mu\text{F}$)

f (Hz)	$ Z_T $ (Ω)	$\angle Z_T$ ($^\circ$)	V_C (Vpp)	ϕ_C ($^\circ$)
100	16595	-73.55	9.591	-16.45
500	5676	-34.11	5.608	-55.89
1000	4962	-18.71	3.207	-71.29
5000	4711	-3.87	0.676	-86.13
10000	4703	-1.94	0.338	-88.06

Table 2: RL circuit, theoretical ($R = 100 \text{ }\Omega$, $L = 33 \text{ mH}$)

f (Hz)	$ Z_T $ (Ω)	$\angle Z_T$ ($^\circ$)	V_L (Vpp)	ϕ_L ($^\circ$)
100	102.1	11.71	2.030	78.29
500	144.0	46.03	7.197	43.97
1000	230.2	64.25	9.007	25.75
5000	1041.5	84.49	9.954	5.51
10000	2075.9	87.24	9.988	2.76

Table 3: RLC circuit, theoretical ($R = 1 \text{ k}\Omega$, $C = 0.33 \text{ }\mu\text{F}$, $L = 10 \text{ mH}$)

f (Hz)	$ Z_T $ (Ω)	$\angle Z_T$ ($^\circ$)	V_C (Vpp)	ϕ_C ($^\circ$)	V_L (Vpp)	ϕ_L ($^\circ$)
100	4919	-78.27	9.804	-11.73	0.013	168.27
500	1368	-43.02	7.052	-46.98	0.230	133.02
1000	1084	-22.76	4.447	-67.24	0.579	112.76
5000	1023	12.28	0.942	-102.28	3.070	77.72
10000	1156	30.12	0.417	-120.12	5.435	59.88

Measurements

Measurements were taken on a Global Specialties PB-503A trainer (the course handout specifies an XK700) using a Tektronix TDS 1002 oscilloscope. The TDS 1002 has no automatic phase measurement, so every phase angle was obtained from time cursors: the period T was read between successive rising zero crossings of the source on channel 1, the shift Δt was read between the rising

zero crossing of the source and the nearest rising zero crossing of the element voltage on channel 2, and the angle was computed as

$$\phi = \pm 360^\circ \frac{\Delta t}{T}.$$

Taking T from the trace rather than from the generator dial removes the generator's frequency accuracy from the result. The sign was assigned from which trace crossed first; a negative angle means the element voltage lags the source. Both channels were DC coupled and their zero references were aligned to the centre graticule line before each set of readings.

Two departures from the ideal model were identified during the lab and are carried through the tables below. First, the PB-503A function generator has an output impedance of roughly $600\,\Omega$ rather than the $50\,\Omega$ assumed for a bench generator, so it could not hold $10\,\text{V}$ peak-to-peak into the low-impedance RL circuit. Fitting the measured source amplitude at node A against a divider model over the five RL test frequencies gives $R_g \approx 575\,\Omega$ with an rms residual of $0.07\,\text{V}$. Because channel 1 was probed at the circuit node and not at the generator terminals, R_g falls outside the measured loop and affects only the absolute amplitudes, not the voltage ratios or the phase angles. Second, the $33\,\text{mH}$ inductor carries a winding resistance of approximately $43\,\Omega$, which is comparable to $|Z_L| = 20.7\,\Omega$ at $100\,\text{Hz}$. A DCR column is therefore included alongside the ideal calculation wherever it changes the prediction appreciably.

Phase data

Table 4: Cursor readings and computed phase angles

Circuit	f (Hz)	T	Δt	ϕ ($^\circ$)
RC, V_C	100	10 ms	900 μs	-32.40
	500	2 ms	360 μs	-64.80
	1000	1 ms	240 μs	-86.40
	5000	200 μs	48 μs	-86.40
	10000	100 μs	24 μs	-86.40
RL, V_L	100	10 ms	480 μs	17.28
	500	2 ms	160 μs	28.80
	1000	1 ms	76 μs	27.36
	5000	200 μs	10 μs	18.00
	10000	100 μs	800 ns	2.88
RLC, V_C	100	10 ms	520 μs	-18.72
	500	2 ms	300 μs	-54.00
	1000	1 ms	200 μs	-72.00
	5000	200 μs	144 μs	-100.80 ^a
	10000	100 μs	68 μs	-115.20 ^a
RLC, V_L	100	10 ms	480 μs	17.28
	500	2 ms	320 μs	57.60
	1000	1 ms	212 μs	76.32
	5000	200 μs	37 μs	66.60
	10000	100 μs	17 μs	61.20

^aAbove resonance ϕ_C passes -90° , and the cursor was placed on the following source crossing rather than the preceding one. The raw ratios give 259.20° and 244.80° ; subtracting 360° recovers the values shown.

Series RC

Table 5: Measured values — series RC ($R = 4.7 \text{ k}\Omega$, $C = 0.1 \text{ }\mu\text{F}$)

f (Hz)	V_C (Vpp)		ϕ_C ($^\circ$)		
	Calculated	Measured	Calculated	Measured	Difference
100	9.591	9.600	-16.45	-32.40	-15.95
500	5.608	5.600	-55.89	-64.80	-8.91
1000	3.207	3.600	-71.29	-86.40	-15.11
5000	0.676	1.400	-86.13	-86.40	-0.27
10000	0.338	0.064 ^b	-88.06	-86.40	1.66

^bRecorded value; the monotonic trend of the other four points suggests 0.64 V peak-to-peak was intended.

The two lowest frequencies agree with the calculation to better than 0.2% in amplitude. The three highest phase readings are identical at -86.40° , corresponding to $\Delta t/T = 0.24$ in all three cases, which is correct at 5 kHz and 10 kHz but not at 1 kHz, where the calculation calls for $\Delta t = 198 \text{ }\mu\text{s}$.

Series RL

The source amplitude at node A was recorded separately at each frequency because the generator could not hold 10 V peak-to-peak into this circuit. The divider model uses $R_g = 600 \text{ }\Omega$ and an open-circuit amplitude of 10 V peak-to-peak.

Table 6: Source amplitude at the circuit node — series RL

f (Hz)	$ Z_{\text{circuit}} $ (Ω)	V_s predicted (Vpp)	V_s measured (Vpp)
100	144.5	1.94	2.08
500	176.6	2.35	2.50
1000	251.9	3.27	3.40
5000	1046.5	8.21	8.20
10000	2078.1	9.44	9.40

Because the source amplitude varies with frequency, the amplitude comparison is made on the ratio V_L/V_s , which is independent of R_g .

Table 7: Measured values — series RL ($R = 100 \text{ }\Omega$, $L = 33 \text{ mH}$)

f (Hz)	measured (Vpp)		V_L/V_s			ϕ_L ($^\circ$)		
	V_s	V_L	Ideal	DCR	Meas.	Ideal	DCR	Meas.
100	2.08	0.800	0.203	0.330	0.385	78.29	17.49	17.28
500	2.50	1.600	0.720	0.635	0.640	43.97	31.53	28.80
1000	3.40	2.030 ^c	0.901	0.841	0.597	25.75	22.88	27.36
5000	8.20	7.800	0.995	0.991	0.951	5.51	5.48	18.00
10000	9.40	9.300	0.999	0.998	0.989	2.76	2.76	2.88

^cThis ratio is lower than the one recorded at 500 Hz. For a series RL circuit V_L/V_s increases monotonically with frequency, so one of the two entries in this row is in error.

The ideal column and the measured column disagree by as much as 61° at 100 Hz, while the column that includes $43\,\Omega$ of winding resistance agrees to 0.2° . The same holds for the amplitude ratio at 500 Hz (0.635 predicted against 0.640 measured) and for the phase at 10 kHz. The 5 kHz phase point is the exception: a $10\,\mu\text{s}$ cursor reading against a $200\,\mu\text{s}$ period is too coarse to resolve an angle whose true value is near $3\,\mu\text{s}$.

Series RLC

Table 8: Measured values — series RLC ($R = 1\,\text{k}\Omega$, $C = 0.33\,\mu\text{F}$, $L = 10\,\text{mH}$)

f (Hz)	V_C (Vpp)		ϕ_C ($^\circ$)		V_L (Vpp)			ϕ_L ($^\circ$)	
	Calc.	Meas.	Calc.	Meas.	Ideal	DCR	Meas.	DCR	Meas.
100	9.804	9.800	-11.73	-18.72	0.013	0.088	0.284	86.09	17.28
500	7.052	7.000	-46.98	-54.00	0.230	0.381	0.388	77.97	57.60
1000	4.447	4.400	-67.24	-72.00	0.579	0.677	0.680	77.52	76.32
5000	0.942	0.900	-102.28	-100.80	3.070	2.976	2.800	70.42	66.60
10000	0.417	0.440	-120.12	-115.20	5.435	5.277	4.860	57.00	61.20

The capacitor amplitudes match the calculation at all five frequencies, the largest deviation being 5.5 % at 10 kHz. The capacitor phase readings above resonance confirm the sign reversal of $\angle Z_T$: the measured -100.80° at 5 kHz and -115.20° at 10 kHz sit within 5° of the calculated values, and both lie beyond -90° , which is only possible once the circuit has become inductive. The inductor amplitudes again track the DCR column rather than the ideal one, most clearly at 1 kHz where the measurement reads 0.680 V against 0.677 V predicted with winding resistance and 0.579 V predicted without it. The inductor phase readings at 100 Hz and 500 Hz do not reconcile with either model; the 100 Hz entry duplicates the value recorded for the RL circuit at the same frequency and is likely a transcription error.

Observations & Discussion

For the most part our measured values matched the theoretical calculations, but the agreement was not uniform across the three circuits and the places where it broke down turned out to be more instructive than the places where it held. The RC circuit tracked theory the closest. The capacitor voltage at 100 Hz and 500 Hz came out within a couple tenths of a percent of the calculated amplitude, and the phase at the two lowest frequencies was within about 9° . The one problem in the RC data is that the three highest phase readings all came out to exactly -86.40° , which corresponds to $\Delta t/T = 0.24$ in every case. That value is right at 5 kHz and 10 kHz, but at 1 kHz the calculation calls for $\Delta t = 198\,\mu\text{s}$ and a phase near -71° , so getting -86° there means the cursor most likely was not moved between those measurements. It is a reading error rather than anything the circuit did.

The RL circuit is where the ideal model and the measurements disagreed the most, and both major sources of error in this lab showed up in it. The first is the function generator. The PB-503A does not hold 10 V peak-to-peak into a low-impedance load; at 100 Hz the source at the circuit node had sagged to about 2 V peak-to-peak. Fitting the measured source amplitude against a voltage-divider model across all five RL frequencies gives an output impedance of roughly $600\,\Omega$, not the $50\,\Omega$ a bench generator is usually assumed to have, with an rms residual of only 0.07 V. Because we probed channel 1 at the circuit node rather than at the generator terminals, this $600\,\Omega$ sits outside

the loop we measured and does not affect the voltage ratios or the phase, which is why we compared V_L/V_s rather than V_L alone. The second source is the inductor itself. An ideal 33 mH inductor has $|Z_L| = 20.7\ \Omega$ at 100 Hz, but a real inductor also has winding resistance, and ours measured on the order of $43\ \Omega$. At low frequency that resistance is larger than the inductive reactance, so it dominates the element voltage. The effect on the phase is dramatic: the ideal calculation predicts $\phi_L = 78.3^\circ$ at 100 Hz, our measurement gave 17.3° , and a model that includes the $43\ \Omega$ of series resistance predicts 17.5° . The measured value sits within 0.2° of the corrected model, so the 61° discrepancy is not error at all; it is the winding resistance, which the ideal formula simply leaves out. The same correction brings the amplitude ratios into line at the low-frequency end.

As we varied the frequency, the measured voltages moved exactly the way the impedances predict. In the RC circuit the capacitor voltage started high and fell as frequency rose, because $|Z_C| = 1/\omega C$ shrinks while the resistor stays fixed, so a smaller and smaller share of the source appears across the capacitor. This is low-pass behavior, and the phase confirmed it, moving from near 0° at 100 Hz toward -90° at the high end as the circuit became almost purely resistive from the capacitor's point of view. The RL circuit did the opposite. The inductor voltage started small and climbed toward the full source voltage as frequency rose, because $|Z_L| = \omega L$ grows with frequency, so more of the source drops across the inductor. That is high-pass behavior, and its phase moved from a large positive angle toward 0° as the inductor came to dominate the loop. The two reactive elements are mirror images of each other, which is the whole point of running the same sweep on both.

The RLC circuit was the most interesting because it does something neither of the single-reactance circuits can. Its resonant frequency is $f_0 = 1/(2\pi\sqrt{LC}) = 2.77\text{ kHz}$, which falls between our 1 kHz and 5 kHz test points, so the total reactance changes sign as the sweep passes through it: the circuit is capacitive below f_0 and inductive above it. Our measured capacitor phase caught this crossover. Below resonance ϕ_C stayed between 0° and -90° , but at 5 kHz and 10 kHz it read -100.8° and -115.2° , both beyond -90° , which can only happen once the net reactance has gone inductive and pushed the current phase past the point a purely capacitive divider could reach. Those two points also illustrate the wrap problem with the cursor method: the raw ratios gave 259.2° and 244.8° because we placed the cursor on the source crossing after the element crossing rather than before it, and subtracting 360° recovers the correct values. The inductor amplitudes in this circuit again followed the model that includes winding resistance rather than the ideal one, most clearly at 1 kHz where we measured 0.680 V against 0.677 V predicted with resistance and 0.579 V without it. The low-frequency inductor phase readings in the RLC circuit did not reconcile with either model, and the 100 Hz entry is identical to the RL circuit's 100 Hz value, which points to a transcription slip in the notebook rather than a real measurement.

The output waveforms themselves did not change shape as we swept the frequency. They stayed sinusoidal at the same frequency as the source throughout, which is what we expect from linear R, L, and C elements: a sinusoid driven into a linear network comes out a sinusoid at the same frequency, changed only in amplitude and phase. What varied was the amplitude of the element voltage and its phase relative to the source, and, in the RL circuit, the amplitude of the source itself once the generator's output impedance came into play. We saw no clipping or distortion, which tells us the generator was never driven into overload even when the RL load pulled its output down to a couple of volts.

Overall the results confirmed the main ideas of the lab. Impedance behaves like resistance in that the series and divider rules carry straight over, but unlike resistance it is complex and frequency-dependent, so the element voltages come out shifted in phase and the divide between them shifts as the frequency changes. Where our numbers diverged from the ideal calculation, the difference was almost always one of two real effects the ideal model leaves out — the generator's

$600\,\Omega$ output impedance and the inductor's $43\,\Omega$ winding resistance — and once those are folded in, the measurements line up with the theory.